

Loss Geometry and Scaling Breaks Reveal How MLP Architectures Generalize

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Abstract—We study how architectural depth shapes loss-landscape geometry and generalization. For multilayer perceptrons of depths $1 - 10$ (25 restarts each), we measure Euclidean descriptors (RMS radius, Hessian curvature) and information-geometric descriptors (Fisher–Rao curvature and a trace-normalized Fisher chord sensitivity), all computed from training data only. The best-performing depths ($d = 5-6$) exhibit a dual signature: low Euclidean curvature and RMS radius, yet high Fisher–Rao curvature and sensitivity. When plotted against *validation* (but not training) loss, all descriptors collapse onto an almost one-dimensional trajectory indexed by depth, indicating that geometry encodes out-of-sample behavior. Moreover, the ratio of Hessian and Fisher–Rao curvature follows a broken scaling law with validation loss: two distinct power-law regimes separated by a break point immediately preceding the optimal depths (exponents ≈ 2 pre-optimal and ≈ 5 post-optimal). These findings position depth as a key driver of geometry, reveal a geometric transition at the best architectures, and suggest that a small set of trace-normalized, Fisher-aware metrics can guide architecture selection with reduced dependence on held-out data.

Index Terms—loss landscape geometry, generalization, neural network architecture, Fisher–Rao metric, curvature, flat minima, information geometry, model selection, overparameterization

I. INTRODUCTION

Data, compute, and architecture remain core bottlenecks in modern machine learning. A common strategy to mitigate compute and data costs is to study small-scale settings and extrapolate via neural scaling laws [1], where test loss empirically follows power-law trends with respect to model size, dataset size, or compute budget. Yet, in practice, large held-out sets are typically reserved to select models and prevent overfitting; in data-scarce regimes this reduces the effective training set and can yield unstable estimates [2], [3]. If we could replace (or substantially shrink) held-out validation with geometry-based indicators computed from training alone, we could better utilize limited data while still guiding model selection.

At the same time, deep networks routinely generalize well despite extreme overparameterization, contradicting classical capacity-based expectations (e.g., VC-dimension) that predict severe overfitting without strong regularization [4]. This gap has motivated a geometric perspective on generalization that examines properties of the loss surface. Empirically, flatter

minima tend to generalize better [5], small-batch methods often find broader basins than large-batch training [6], and systematic benchmarks place curvature-based measures among the strongest correlates of test error [7]. However, sharpness notions can be sensitive to reparameterization [8], and recent PAC-Bayesian work reframes these effects in bounds that involve curvature and gradient terms [9].

Information geometry provides a principled lens for these observations. For models trained with cross-entropy, the expected Hessian coincides with the Fisher information, tying curvature to the intrinsic statistical geometry of the predictor [10], [11]. This connection underlies natural-gradient methods that rescale updates by the inverse Fisher to improve optimization and theory [12], [13]. Despite this progress, the role of *architecture*—in particular, depth—in shaping loss-landscape geometry and, in turn, generalization remains incompletely understood. Even for multilayer perceptrons (MLPs), test performance varies non-monotonically with depth and is often attributed to representational compression or “tunnel” effects [14], but the accompanying geometric mechanisms are underexplored.

We position our study at the intersection of Fisher–Rao capacity measures [15] and recent work on broken neural scaling laws [16]. Anticipating our results, we show that a scaling break point in geometry space—visible only against validation loss—aligns with the optimal depth, linking these two lines of work.

We achieve this by studying how architectural depth organizes the loss landscape of multilayer perceptrons in both Euclidean and Fisher–Rao geometries, and how this geometry relates to generalization. Across depths and restarts, we observe a distinctive signature for the best-performing architectures: they lie in basins that are flat in Euclidean terms yet sharp in Fisher–Rao terms. When geometric descriptors are plotted against *validation* loss (not training loss), they collapse toward a near one-dimensional trajectory parameterized by depth. The trajectory for the ratio of Euclidean to Fisher curvature obeys a broken scaling law with a data-driven break point immediately preceding the optimal depths. These results indicate that geometry carries information about generalization beyond training loss and suggest a practical route to geometry-aware architecture selection with reduced reliance on large held-out sets.

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II. METHODS

We consider fully connected feed-forward networks (multi-layer perceptrons, MLPs) of varying depth d and fixed hidden-layer width w . Each network defines a map

$$f_{\text{MLP}}^{(d)} : \mathbb{R}^{784} \rightarrow \mathbb{R}^{10},$$

taking a flattened 28×28 MNIST image to a 10-dimensional output. Depths $d \in \{1, 2, \dots, 10\}$ are explored with $w = 48$ nodes per hidden layer, using ReLU activation after each linear transformation.

To limit runtime, MLPs are trained on a subset of MNIST with $n_{\text{train}} = 5000$ examples and validated on $n_{\text{val}} = 1000$ held-out images. Pixel intensities are rescaled to $[0, 1]$, and labels are one-hot encoded. Training and validation losses use the mean-squared error (MSE) objective, e.g.,

$$\ell_{\text{train}}(\theta) = \frac{1}{n_{\text{train}}} \sum_{i=1}^{n_{\text{train}}} \|f_{\theta}(x_i) - y_i\|_2^2,$$

with an analogous definition for $\ell_{\text{val}}(\theta)$.

For each depth d , we perform $R = 25$ independent restarts with $\theta^{(0)} \sim \mathcal{N}(0, \mathbf{I})$ to sample multiple local minima. Parameters θ^* are estimated by Adam (learning rate $\alpha = 10^{-3}$, default momentum), minimizing the loss until $|\ell_{\text{train}}^{(t)} - \ell_{\text{train}}^{(t-1)}| < 10^{-6}$ or 10^5 iterations are reached.

For each depth d and restart i , we record:

- Initial parameters $\theta_{d,i}^0$
- Converged parameters $\theta_{d,i}^*$
- Final training loss $\ell_{\text{train}}(\theta_{d,i}^*)$
- Validation loss $\ell_{\text{val}}(\theta_{d,i}^*)$

The resulting collections of 25 restarts for each of the ten depths d form the dataset for all subsequent analyses.

III. GEOMETRIC DESCRIPTORS

A. Euclidean Geometry

1) *Hessian Curvature*: We first characterize the local curvature of the loss landscape at converged parameters θ^* . Defining the Hessian $H(\theta^*) = \nabla^2 \ell(\theta^*) \in \mathbb{R}^{p \times p}$, the Frobenius norm curvature is

$$\kappa := \|H(\theta^*)\|_F = \sqrt{\text{tr}(H^\top H)} = \sqrt{\sum_{k=1}^p \lambda_k^2},$$

where $\{\lambda_k\}$ are the eigenvalues of H . This scalar aggregates curvature magnitude across all parameter directions without cancellation between positive and negative values, following prior work on flatness measures [17].

Exact evaluation of κ is $\mathcal{O}(p^3)$; we instead estimate $\text{tr}(H^2)$ using Hutchinson’s stochastic trace estimator [18] with $m = 45$ Rademacher probes, yielding a relative error of $\mathcal{O}(1/\sqrt{m}) \approx 15\%$ [19]. Each probe requires a Hessian–vector product, which we compute efficiently via reverse–over–forward automatic differentiation, avoiding explicit formation of H .

2) *RMS Radius*: To quantify variability between solutions found from independent initializations, we compute the root–mean–squared (RMS) radius for each depth d . Let $\{\theta_{d,i}\}_{i=1}^{n_r}$ denote the $n_r = 25$ converged parameter vectors, arranged as the rows of $\Theta_d \in \mathbb{R}^{n_r \times p}$. The Gram matrix $\Sigma_d = \Theta_d \Theta_d^\top$ yields

$$\rho_{\text{RMS}}(d) := \sqrt{\frac{\text{tr}(\Sigma_d)}{n_r}},$$

the average Euclidean norm of the minimizers, which directly reflects how tightly restarts cluster in parameter space [20].

B. Information Geometry

We now turn to descriptors derived from the Fisher information metric, which endows parameter space with a Riemannian geometry [10].

1) *Fisher–Rao Curvature*: All Fisher–based quantities in this work are defined for the *cross–entropy* (CE) loss. For parameters $\theta \in \mathbb{R}^p$ and the (empirical) Fisher information matrix $F(\theta) \in \mathbb{R}^{p \times p}$, the Fisher–Rao norm is

$$\|\theta\|_{\text{fr}}^2 = \theta^\top F(\theta) \theta.$$

Because its magnitude depends on both the parameter dimension p and the overall spectral scale of $F(\theta)$, we report a trace–normalized variant

$$\|\tilde{\theta}\|_{\text{fr}}^2 = \frac{\|\theta\|_{\text{fr}}^2}{\text{tr}(F(\theta))/p},$$

which divides by the mean eigenvalue of $F(\theta)$ and thereby removes global spectral scaling, enabling comparisons across depths. We define the Fisher–Rao curvature at the converged solution as

$$\kappa_{\text{fr}} := \|\tilde{\theta}^*\|_{\text{fr}},$$

a dimensionless measure of intrinsic sharpness in Fisher geometry at θ^* .

Under CE, the *expected* Hessian coincides with the Fisher information, motivating Fisher as a curvature notion in the statistical (information–geometric) sense. In practice we use the *empirical* Fisher built from per–example CE gradients and form a low–rank factorization via output Jacobians. For n_s sampled training points, let $J_i \in \mathbb{R}^{C \times p}$ be the Jacobian of the C outputs at x_i ; stacking all rows yields

$$G \in \mathbb{R}^{(n_s C) \times p}, \quad G \approx U_r \Sigma_r V_r^\top,$$

so that

$$F(\theta) \approx V_r (\Sigma_r^2 / n_s) V_r^\top,$$

reducing storage from $\mathcal{O}(p^2)$ to $\mathcal{O}(pr)$ and enabling efficient quadratic–form evaluation (for softmax CE this coincides with the generalized Gauss–Newton matrix, providing a PSD curvature proxy [11], [13]). In our experiments we used rank $r = 100$ with $n_s = 1500$ points per depth.

Empirically, trace normalization substantially improves alignment with validation loss: the global Spearman correlation of the raw FR norm with validation loss is $\rho =$

-0.227 ± 0.140 , which improves to $\rho = -0.825 \pm 0.045$ after normalization (95% CIs estimated via a nonparametric percentile bootstrap with 1000 resamples over all 250 (depth, restart) pairs). The normalization also places κ_{fr} on a comparable scale to the Euclidean curvature summary, facilitating joint visualization and model-to-model comparisons.

2) *Fisher–Rao Sensitivity*: Let $\Delta\theta = \theta^* - \theta^0$ denote the displacement from initialization to the final parameters. We sample m evenly spaced points along the straight Euclidean chord

$$\theta_k = \theta^0 + t_k \Delta\theta, \quad t_k = \frac{k}{m-1}, \quad k = 0, \dots, m-1,$$

and at each point evaluate the Fisher norm of the *fixed* displacement $\Delta\theta$ under the (empirical) Fisher for cross-entropy:

$$c_k = \sqrt{\Delta\theta^\top F(\theta_k) \Delta\theta}.$$

Averaging along the chord gives the (Fisher) chord length

$$d_{\text{fr}} = \frac{1}{m} \sum_{k=0}^{m-1} c_k,$$

which is *not* the Fisher–Rao geodesic distance, but a tractable proxy for the *average* Fisher curvature encountered along the direct path from θ^0 to θ^* .

To compare across depths, we remove global spectral scaling by normalizing with the mean eigenvalue of the *averaged* Fisher along the chord:

$$\tilde{d}_{\text{fr}} = \frac{d_{\text{fr}}}{\sqrt{\text{tr}(F_{\text{avg}})/p}}, \quad F_{\text{avg}} = \frac{1}{m} \sum_{k=0}^{m-1} F(\theta_k).$$

Finally, we define the *Fisher–Rao sensitivity* as a dimensionless ratio that compares Fisher length to Euclidean length per parameter:

$$\varsigma = \frac{\tilde{d}_{\text{fr}}}{\|\Delta\theta\|_2 / \sqrt{p}}.$$

A large ς indicates that the displacement from initialization to the solution lies preferentially along directions with high Fisher curvature—i.e., directions to which the model’s CE predictions are most sensitive—so the training displacement is closer (on average) to a natural-gradient-aligned direction [12].

We use the same empirical Fisher construction as in the curvature subsection (CE loss, low-rank factorization of output Jacobians), reusing a fixed subset of $n_s = 1500$ training points per depth and a truncation rank $r = 100$. We sample $m = 16$ chord points. Trace normalization materially improves agreement with validation loss: the global Spearman correlation for sensitivity improves from $\rho = -0.587 \pm 0.095$ (unnormalized) to $\rho = -0.880 \pm 0.033$ (trace-normalized) (95% CIs estimated via a nonparametric percentile bootstrap with 1000 resamples over all 250 (depth, restart) pairs). This places ς on a comparable scale across depths and reduces variance due to Fisher spectrum rescaling.

IV. RESULTS

A. Loss Landscape Geometry of Optimal Architectures

The best-performing depths are $d = 5$ and $d = 6$, which achieve the lowest validation loss. Figure 1 presents geometric descriptors as functions of depth and as scatter plots versus *log* training and validation loss, with the optimal architectures outlined in black.

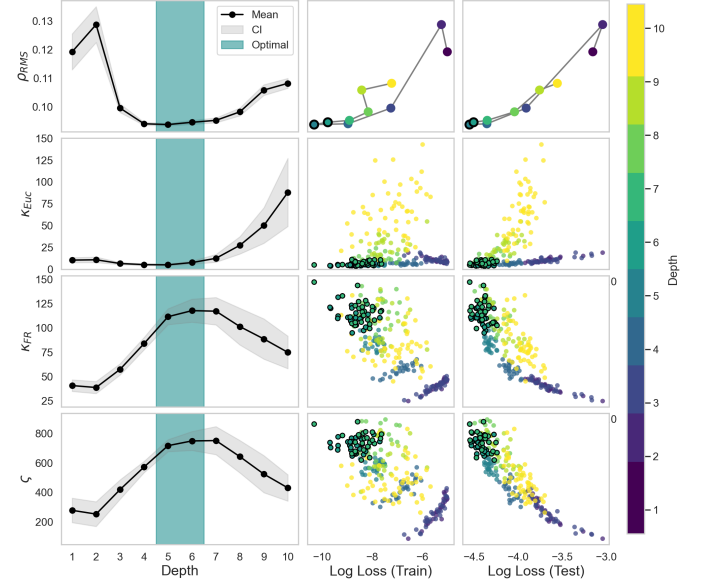


Fig. 1. Geometric descriptors across MLP depths, highlighting the controlled-complexity regime ($d \leq 6$) and the complexity-dominated regime ($d > 6$). Left: mean $\pm$ one standard deviation of each metric over 25 initializations. Middle: metrics versus training log-loss. Right: metrics versus validation log-loss. Black outlines mark the optimal depths $d = 5$ and $d = 6$, characterized by low Euclidean curvature and RMS radius but high Fisher–Rao curvature and sensitivity.

Across all descriptors—RMS radius (ρ), Euclidean curvature ($\kappa_{\text{Euc}} := \|H(\theta^*)\|_F$), Fisher–Rao curvature ($\kappa_{\text{FR}} := \|\tilde{\theta}^*\|_{\text{fr}}$), and Fisher–Rao sensitivity (ς), optimal depths $d = 5, 6$ exhibit a distinctive signature: in Euclidean terms they minimize ρ and κ_{Euc} (compact weights and flat basins), while in Fisher terms they maximize κ_{FR} and ς (large curvature and alignment of training along prediction-sensitive directions). Although these descriptors are computed using training data, they organize far more cleanly when plotted against validation loss than against training loss, collapsing onto a near one-dimensional trajectory parameterized by depth.

The depth profiles indicate a regime change centered on the optimal depths. For $d < 5$, κ_{Euc} scales approximately linearly with log validation loss, while κ_{FR} rises more steeply; beyond $d = 5$, κ_{Euc} grows superlinearly relative to loss, whereas κ_{FR} tends back toward proportional scaling. Fisher–Rao sensitivity tracks the same transition, and ρ achieves its minimum at the same depths, reinforcing that the shift is geometric rather than an artifact of any single descriptor.

B. Geometric Broken Scaling Laws

To quantify the regime change, we fitted two-segment broken power laws between validation loss (x) and each descriptor (y). The split (break) point parameter s defines two regimes, each getting their own power law:

$$\{(x_i, y_i) : d_i < s\} \quad \text{and} \quad \{(x_i, y_i) : d_i > s\}.$$

Here, the regression is blind to depth except for the partition. We search s over admissible splits (half-integers to indicate boundaries, e.g., $s = 4.5$) and select the s that maximizes mean Pearson r across the two segments, or equivalently minimizes total squared error. Confidence intervals for the exponents are obtained by nonparametric bootstrap over runs within each regime (resampling restarts at fixed depths), refitting in log-log space to form percentile intervals. This is a geometry-space analogue of broken scaling laws used for data/compute/model-size scaling [16], adapted here so that the split is indexed by architecture depth while each segment’s fit uses only (x, y) . Figure 2 shows the results of this process for different potential metrics.

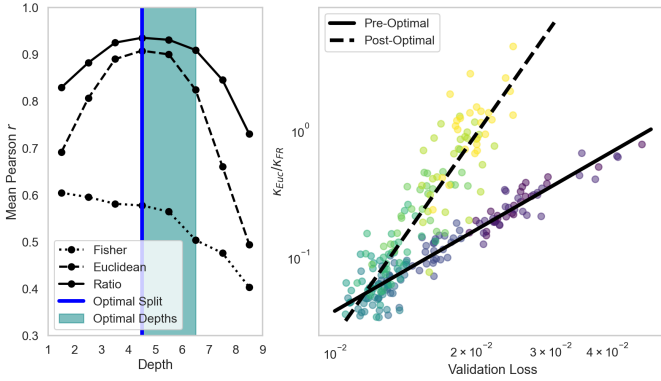


Fig. 2. Broken scaling law with validation loss. Left: break point location and correlation strength for Euclidean curvature κ_{Euc} , Fisher–Rao curvature κ_{FR} , and their ratio $\kappa_{\text{Euc}}/\kappa_{\text{FR}}$. Optimal depths ($d = 5, 6$) are shaded; the vertical line marks the best-fit break point, just before them. Right: scaling of curvature ratio with validation loss in the pre-optimal ($\alpha = 2.05 \pm 0.13$, $r = 0.955$) and post-optimal ($\alpha = 5.10 \pm 0.36$, $r = 0.916$) regimes, separated at the $d = 4$ – 5 break point.

Between κ_{fr} , κ_{Euc} , and $\kappa_{\text{Euc}}/\kappa_{\text{fr}}$, the cleanest broken-scaling behavior occurs for the curvature ratio $\kappa_{\text{Euc}}/\kappa_{\text{FR}}$. A single break point selected by correlation maximization lies at $s = 4.5$, immediately preceding the best depths. The pre- and post-break fits show strong linearity on the log-log plot (Pearson $r = 0.955$ and $r = 0.916$), with scaling exponents $\alpha_{\text{pre}} = 2.05 \pm 0.13$ and $\alpha_{\text{post}} = 5.10 \pm 0.36$ (95% CIs).

V. DISCUSSION

The results show that training-loss-derived geometric descriptors— ρ , κ_{Euc} , κ_{FR} , and ζ —align tightly with validation performance despite using no validation information. When plotted against validation loss, these quantities collapse to an almost one-dimensional trajectory indexed by depth,

whereas their relationships with training loss remain more dispersed. At the best depths ($d = 5, 6$), the minima are Euclidean-flat yet Fisher-sharp: basins are broad and robust in weight space while curvature is concentrated along statistically meaningful directions. This reconciles Euclidean flatness heuristics with information-geometric capacity arguments grounded in the Fisher–Rao metric and natural-gradient invariances [15].

A second finding is a broken power law in *geometry space*. The curvature ratio $\kappa_{\text{Euc}}/\kappa_{\text{FR}}$ versus validation loss exhibits two distinct power-law regimes with a break point immediately preceding the optimal depths; the scaling exponent increases from roughly $\alpha \approx 2$ pre-optimal to $\alpha \approx 5$ post-optimal. Importantly, the metric that exhibited the strongest fit to a broken power law used *both* Fisher–Rao and Euclidean geometric features.

Geometrically, before the break, validation loss can be reduced without rapidly steepening the Euclidean basin relative to Fisher curvature; beyond the break, additional depth yields diminishing returns, demanding disproportionately larger increases in Euclidean curvature for comparable gains in validation performance. The fact that this behavior appears clear against validation loss (not training loss) suggests a change in the *generalization* mechanism rather than in optimization dynamics. Conceptually, our results marry the invariant geometric capacity view of Liang *et al.* with the broken-scaling perspective of Caballero *et al.*: the Fisher–Rao lens provides reparameterization-aware coordinates in which a clear scaling break point emerges [15], [16]. Practically, locating this geometric break point (e.g., by monitoring $\kappa_{\text{Euc}}/\kappa_{\text{FR}}$ across depth) offers a simple heuristic for architecture selection without exhaustive validation sweeps. A natural next test is to move beyond depth in MLPs and evaluate, in settings from geometric deep learning (GDL) and topological data analysis (TDA), whether architectures that *align* with the topology and symmetries of the data [21], [22] *systematically induce* the same “good-geometry” signature we observe here: low Euclidean curvature and RMS radius at the solution together with high Fisher–Rao curvature and chord sensitivity, plus a clean broken scaling of the curvature ratio versus validation loss. This hypothesis is falsifiable: if architectural biases are misaligned with the data’s structure, one should *not* see the Euclidean-flat / Fisher-sharp signature nor the same breakpoint behavior (cf. architectural blind spots in [23]). Physics-informed architectures provide another controlled setting where known invariances and constraints can be injected [24]; TDA-based probes of representation geometry could independently verify whether alignment improves generalization without test data [25]. Concretely, the proposed experiment is to hold a dataset fixed while varying the inductive bias (e.g., equivariant vs. non-equivariant, graph vs. MLP), and to hold the architecture fixed while varying synthetic data topology, then check whether the alignment condition predicts the emergence of the Euclidean-flat / Fisher-sharp signature and the same geometric breakpoint. This would extend our depth-only study to richer architectural axes and provide a precise, geometry-based test of GDL/TDA effectiveness.

Limitations include the use of a single dataset and architecture family and a low-rank empirical Fisher with chord-based (non-geodesic) averaging. Replication across datasets and broader architectural axes (width, residual/skip connections, equivariant and graph networks) is an immediate next step, along with alternative Fisher estimators and path metrics. It will also be important to test whether the broken scaling exponents are universal or task dependent and whether similar geometry-only break points arise when scaling other aspects of architecture. Nevertheless, the consistent Euclidean–Fisher decoupling at the optimum and the sharp broken law behavior against validation loss indicate a robust coupling between architecture, data geometry, and generalization that can be exploited for principled, validation-lean model selection.

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